

Unit 3:

Metabolomics and Lipidomics:

Metabolism is the set of life-sustaining chemical reactions in organisms. The main purposes of metabolism are to: convert food or fuel to energy so that cellular processes can run; convert food to building blocks for proteins, lipids, nucleic acids, and carbohydrates; and eliminate metabolic wastes.

These enzyme-catalyzed reactions allow organisms to grow, reproduce, maintain their structures, and respond to their environment.

Metabolomics is a new branch of research to understand metabolism in the 21st century. It focuses on the high-throughput identification and quantification of small metabolites in the biological sample. It explores the study of the metabolome.

Metabolomics is the identification of small molecules known as metabolites.¹ Metabolomics is a powerful companion to other commonly used omics techniques: genomics, transcriptomics, and proteomics. While these techniques provide information about genetic and functional potential, metabolomics goes a layer deeper by providing phenotypic information. Our genomes, transcriptomes, proteomes, and metabolomes don't exist in a vacuum—they impact and are impacted by one another as well as by the outside environment.

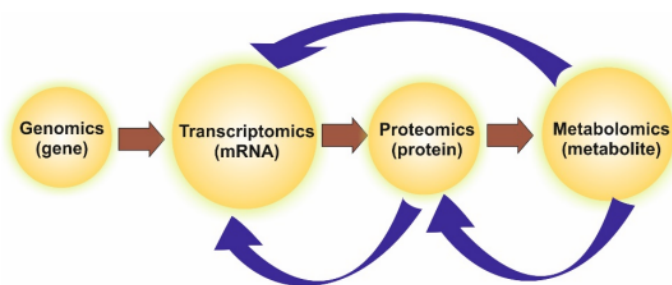


Fig : Inter-relationship of OMICS branches

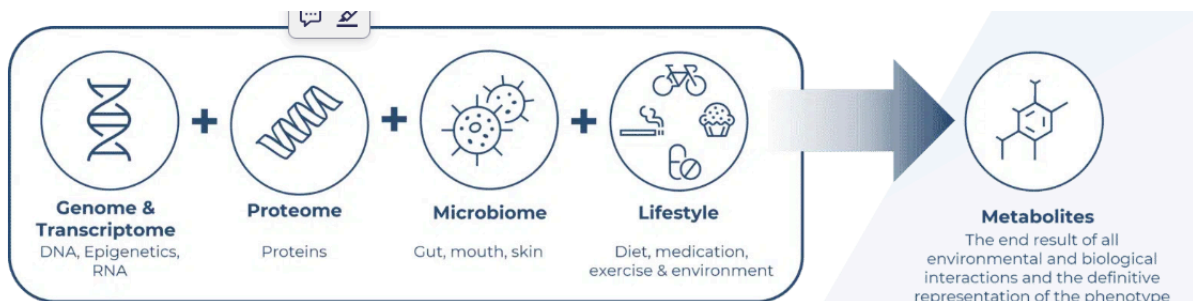


Figure: The complex interrelationship between genes, environment, and functional capacity. The genome, studied via genomics, provides all of the information necessary for creating a functional organism. The transcriptome, studied via transcriptomics, reveals which genes are actually being turned into transcripts as well as which non-coding RNAs are present and can elucidate

effects on the genome or even on the proteome and metabolome. The proteome, studied via proteomics, is the collection of proteins that have actually been translated from RNA transcripts. Finally, the metabolome, studied via metabolomics, includes all small molecules included that are present in an organism under a certain set of conditions. While each of these molecules can influence any of the others, environmental factors, such as therapeutic treatments, can also critically impact gene expression, regulation, protein folding, and metabolic function.

Metabolites are small molecules less than 1.5 kDa in size. The end products of metabolism, metabolites have a wide range of functions, including cell growth support, defense and inhibition, and stimulation. They include amino acids, alcohols, vitamins, polyols, organic acids, and many other types of molecules, and are often the building blocks for larger compounds. Identifying metabolites and how they interact with one another—and how those interactions change under certain conditions—can aid in the discovery and understanding of how organisms work, why diseases develop (or not), why treatments are successful or not, and so much more.

The metabolome is the collection of all metabolites present in an organism. Metabolomes are incredibly fluid and can change drastically with differences in the outside environment, such as nutrient availability, medications, etc. Importantly, the metabolome doesn't just include metabolites produced natively by an organism, but also metabolites produced by the microbiome residing in and on that organism. Microbial metabolites, for example, can have a significant impact on the metabolism of therapeutics inside the host and can impact whether medications such as chemotherapy are effective or not.

Metabolomics is the study of endogenous metabolites, called the metabolome, which is a collection of low molecular weight (50 to 1,500 Da) compounds with a wide range of physiochemical properties. Measuring the metabolome provides important information about biological phenotype and functional status of biochemical pathways. Its proximity to the phenotype of an organism provides complementary information to genomics and proteomics. Metabolites are important modulators, substrates, byproducts, and building blocks for many different biological processes. The presence or absence of specific metabolites in a cell, cell culture, or biofluid provides important information about the physiological and functional status of the biological system. For example, the accumulation of specific metabolite(s) could indicate a defect in the signaling response of a biological pathway, or perhaps optimization of a biosynthetic pathway.

Types of Metabolomics study

It can be divided into targeted and untargeted metabolomics based on the research study .

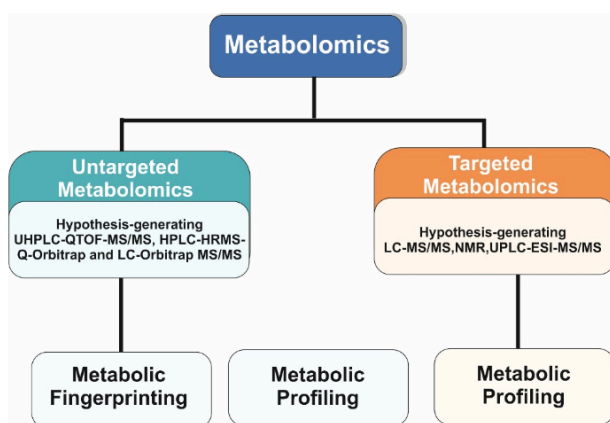
Targeted Metabolomics

Targeted metabolomics deals with the identification and quantification of known metabolites in biological sample such as measuring glucose, pyruvate or ammonia or urea, bilirubin, and cholesterol etc. It is known as hypothesis driven approach. It is important for clinical diagnosis of diseases or study of drug effects. The targeted approach employs triple quadrupole mass spectrometers for identification and quantification for extensive dynamic range of metabolites. The resulting chromatogram is compared with established chemical standards to ensure precise authentication of each metabolite. This approach is very accurate

and repeatable, making it beneficial for clinical diagnostics and biomarker validation. It is useful for metabolite profiling.

Untargeted Metabolomics

Untargeted Metabolomics aims to complete profiling of metabolites without prior knowledge of their identity. It is hypothesis generating approach. It applies to measurement of both known and unknown metabolites across various experimental groups (e.g., healthy versus sick). High-resolution techniques such as LC–MS, GC–MS and NMR spectroscopy are frequently employed for the detection of non-targeted metabolites. The chromatogram data undergo statistical processing in bioinformatics software to find broad view of metabolites and significant metabolites between different groups. It is useful for both metabolic fingerprinting and metabolite profiling.



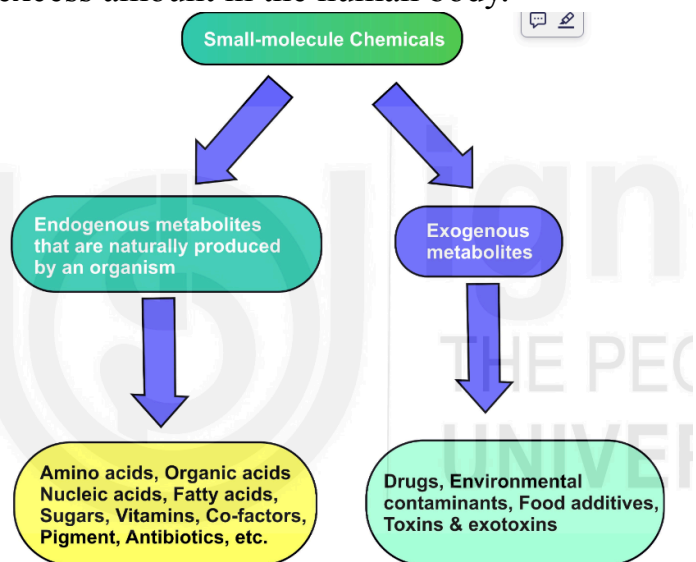
Classification of metabolomics.

There are three categories of metabolites including:

1. **Primary metabolites:** Primary metabolites are generally produced in adequate amount in the body by the metabolic process of the biomolecules i.e. proteins, nucleic acids, lipids and carbohydrates. The primary metabolites are amino acids, nucleotides, sugars, ATP, fatty acids, vitamins, coenzymes etc. which are central molecules for growth, development and reproduction of the living organisms (animal, plants and bacteria). for primarily produced by catabolic process of food stuffs.

2. **Secondary metabolites:** Secondary metabolite are found in plants. In contrast to primary metabolites, secondary metabolites are not directly synthesized by the cellular metabolic processes. Hence, they are secondary metabolites. They are synthesized by secondary metabolism in response to specific environmental conditions. They are heterogeneous group of natural metabolites often found in plant system. They are also known as phytochemicals, including phenolics, flavonoids, terpenes, alkaloids, steroids, tannic acids, pigments (carotenoids) and glycosides. They are regarded as exhibiting pharmacological properties such as antibacterial, anticancer, neuroprotective, hepatoprotective, antidiabetic, and growth- modulating effects, among others, serving as a plant defense mechanism against germs and fungi. They participate in adaptability and signalling pathways.

3. **Xenometabolites:** refers to metabolites of foreign chemicals. They are produced in the body after the consumption of a drug. The drug metabolism enzyme system known as the CYP450 family is responsible for drug metabolism and changed them into either a polar or nonpolar metabolites. Such metabolites can be harmful, useful, or inactive inside the body. In most cases, they are eliminated through the urine and the faeces. For example, CYP450 enzyme converts acetaminophen (known as paracetamol drug) converts into N-acetyl-p-benzoquinone imine (NAPQI), a harmful active metabolite if it is in excess amount in the human body.



Types of metabolites

Consequently, the concentration of metabolites varies among different species, including animals, plants, and microorganisms. Metabolomics research concentrates on the profiling of endogenous compounds, including peptides, oligopeptides, nucleotides, signalling molecules, sugars, secondary metabolites, hormones, neurotransmitters, as well as exogenous components such as drugs, allergens, microbial metabolites, viral particles, pollutants, and drug metabolites.

Case Study:

Liver is the main organ where maximum cellular metabolism occurs
Liver damage in chronic alcoholism disrupts normal metabolic pathways, producing characteristic metabolomic changes.

Carbohydrate metabolism: Impaired gluconeogenesis and glycogen storage alter glucose and lactate levels.

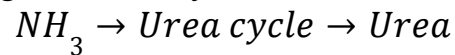
Lipid metabolism: Increased NADH from alcohol metabolism promotes fatty acid synthesis causing fatty liver, elevated triglycerides and ketone bodies.

Protein/amino acid metabolism: Reduced amino acid processing causes increased aromatic amino acids and elevated ammonia.

Bilirubin and bile acid metabolism: Impaired bilirubin conjugation and excretion lead to jaundice and increased bilirubin.

Oxidative stress: Alcohol-induced reactive oxygen species decrease glutathione and increase lipid peroxidation products.

Ammonia accumulates because the damaged liver cannot effectively convert ammonia into urea through the urea cycle.



Hepatocyte injury decreases urea cycle activity.

Portal-systemic shunting allows ammonia to bypass the liver.

Increased intestinal ammonia production also contributes.

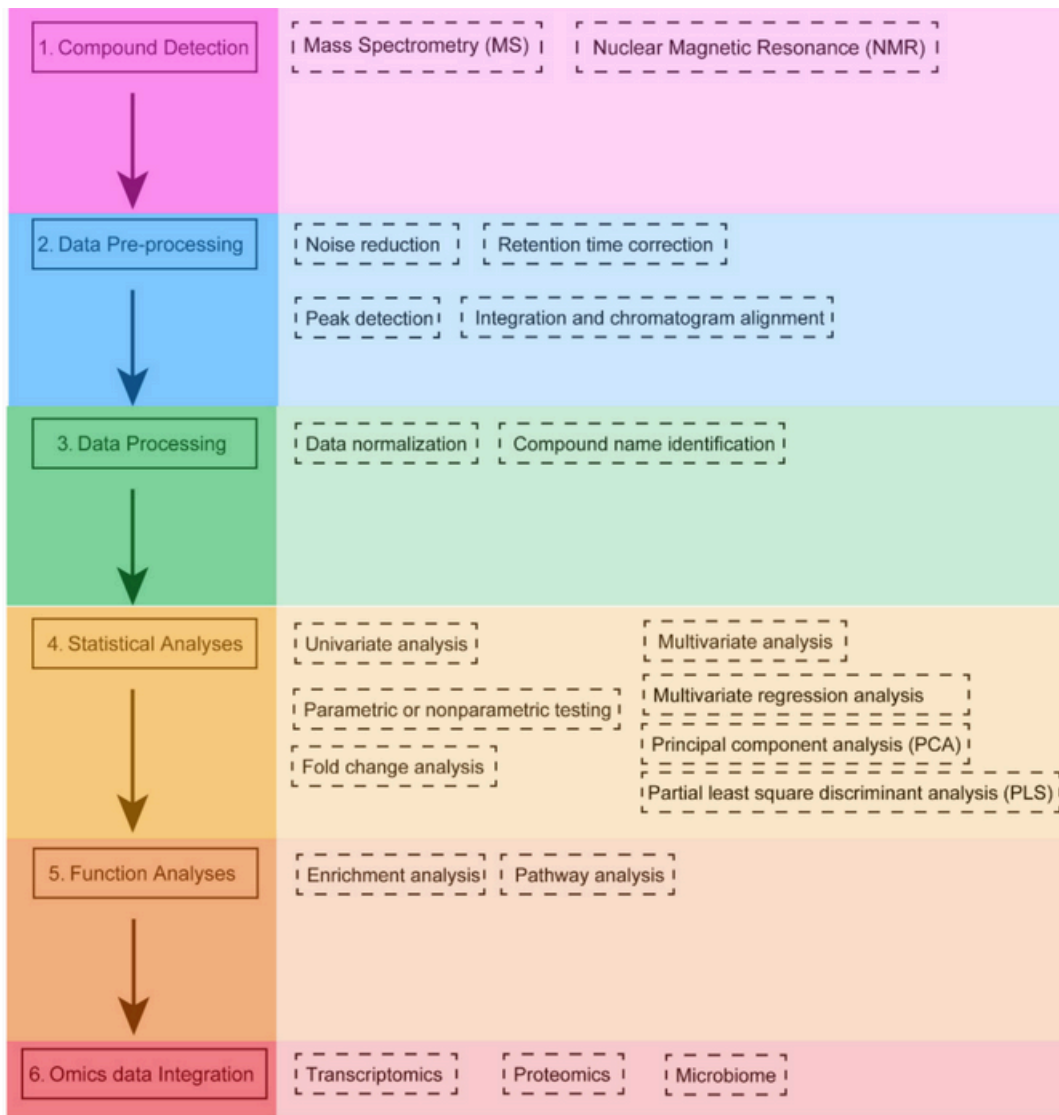
Metabolomics helps analyze metabolic changes occurring in liver disease.

Importance:

- Detects liver disease at an early stage.
- Identifies specific biomarkers such as bile acids, amino acids, and lipids.
- Differentiates alcoholic liver disease, hepatitis, fatty liver, and cirrhosis.
- Monitors disease progression and treatment response.
- Supports personalized treatment strategies.

Measuring the metabolome

The metabolome is the final downstream product of the genome and is defined as the total quantitative collection of small molecular weight compounds (metabolites) present in a cell or organism which participate in metabolic reactions required for growth, maintenance and normal function. The estimated size of the metabolome is large (*S. cerevisiae* approximately 600 metabolites, plant kingdom up to 200 000 metabolites, though significantly lower numbers for different species, and analysis of the human metabolome reveals even greater complexity and number of metabolites detected). In comparison to the proteome or transcriptome, the metabolome is more diverse in chemical and physical properties because of the larger variations in atomic arrangements. Studies of the metabolome include the analysis of a wide range of chemical species, from low molecular weight polar volatiles such as ethanol, to high molecular weight polar glucosides, non-polar lipids and inorganic species.⁸ The range of metabolite concentrations can vary over nine orders of magnitude (pM–mM). The metabolomics experiment (or metabolomics pipeline²⁹) is composed of different stages (experimental design, sampling, sample preparation, sample analysis, data pre-processing and data processing)



Typical workflow of metabolomics analysis. Metabolites are detected by using specific detection techniques (compound detection). Raw signals are then pre-processed to produce data in a suitable format for subsequent statistical analysis (data pre-processing). Then, data normalization is used to reduce the system and technical bias. For untargeted studies, metabolites are identified from spectral information in some given database (data processing). Univariate and multivariate statistical analyses are used to identify significantly expressed metabolites (statistical analyses). Next, the significantly expressed metabolites are subsequently linked to the biological context by using enrichment and pathway analysis (function analyses). Finally, metabolomics data may be integrated with other omics data (transcriptomics, proteomics, or the microbiome) to gain a comprehensive understanding of the molecular mechanisms of pathophysiological processes.

Software Tools for Metabolomics Data Analysis and Integration

Metabolomics analyses need powerful software tools to address the vast amount and variety of data. Excellent metabolomics software should include one or more of the following functions: (1) the ability to process of raw spectral data, (2) statistical analysis to find significantly expressed metabolites, (3) the ability to connect to metabolite databases for metabolite identification, (4) bioinformatics analysis and visualization of molecular interaction networks, and (5) the ability to integrate and analyze multi-omics data.

Features of several most used metabolomics data analysis tools.

Name	Year	Description
Mzmine3	2022	MZmine3 builds on the success of MZmine 2 with many features focused on improving the user-friendly graphical
MetaboAnalyst 5.0	2021	Comprehensive web-based tool for comprehensive metabolomics data analysis, interpretation, and integration with other omics data.
LipidSig	2021	Web-based tool for lipidomic data analysis
MS-DIAL 4.0	2020	Lipidome atlas in MS-DIAL 4.0
EI-MAVEN	2019	Fast, Robust, and User-Friendly Mass Spectrometry Data Processing Engine for Metabolomics
MetFlow	2019	Interactive and integrated web server for metabolomics data cleaning and differential metabolite discovery.
LION	2019	Web-based ontology enrichment tool for lipidomic data analysis.
Omicsnet	2018	Web-based tool for creation and visual analysis of biological networks in 3D space
METLIN	2018	Technology platform for the identification of known and unknown metabolites and other chemical entities.
PaintOmics 3	2018	Web-based resource for the integrated visualization of multiple omics data types onto KEGG pathway diagrams.
LipuDex	2018	Integrated Software Package for High-Confidence Lipid Identification
LipidMatch	2017	Automated workflow for rule-based lipid identification using untargeted high-resolution tandem mass spectrometry data
3Omics	2013	One-click web tool for fast analysis and visualization of multi-omics data.
IMPaLa	2011	Pathway analysis of transcriptomics or proteomics and metabolomics data.
MetPA	2010	Pathway analysis for metabolomics data.
MassTRIX	2008	Tool for high precision MS data annotation.
MetaCore™	2004	Commercial tool for functional analysis and integrated analysis of multi-omics data.

Lipidomics is the large-scale study of lipids in biological systems. In metabolic syndrome, lipid metabolism becomes dysregulated due to obesity, insulin resistance, and diabetes. By comparing lipid profiles of obese individuals, diabetic patients, and healthy controls, researchers can identify disease-specific lipid changes.

Mass spectrometry–based lipidomics can:

Detect and quantify hundreds to thousands of lipid species simultaneously.

Identify altered lipid classes such as ceramides, phosphatidylcholines, and omega-3 phospholipids.

Reveal patterns associated with insulin resistance, inflammation, and cardiovascular risk.

Discover lipid molecules that consistently differ between healthy and diseased groups, which can serve as biomarkers for early diagnosis or prognosis.

Thus, lipidomics provides sensitive and comprehensive molecular signatures of metabolic syndrome.

Ceramide Signaling plays a major role in metabolic syndrome.

In insulin resistance:

Ceramides interfere with the insulin signaling pathway by inhibiting Akt/protein kinase B, a key molecule required for glucose uptake.

Reduced Akt activity decreases glucose transport into muscle and adipose tissues, leading to elevated blood glucose levels.

Ceramides also promote lipid accumulation in tissues, worsening metabolic dysfunction.

In inflammation:

Ceramides activate inflammatory pathways such as NF-κB and cytokine production.

They increase oxidative stress and mitochondrial dysfunction.

Chronic elevation of ceramides contributes to low-grade systemic inflammation commonly observed in obesity and type 2 diabetes.

Therefore, elevated ceramide levels are strongly associated with both insulin resistance and inflammatory progression in metabolic syndrome.

A typical lipidomics workflow includes the following steps:

Sample collection and storage

Plasma samples are collected from subjects under standardized conditions.

Samples are stored at low temperatures (−80°C) to prevent lipid degradation.

Lipid extraction

Lipids are isolated using organic solvent extraction methods such as Folch or Bligh–Dyer extraction.

Mass spectrometry analysis

Lipids are separated using liquid chromatography (LC) and analyzed by mass spectrometry (MS/MS).

The instrument detects lipid mass-to-charge ratios and fragmentation patterns.

Data processing

Specialized software identifies and quantifies lipid species.

Statistical analysis compares lipid abundance among healthy and diseased groups.

Biomarker identification

Significant lipid alterations are validated and correlated with clinical parameters such as glucose levels or BMI.

Identified lipid biomarkers can be translated into clinical diagnostics through several steps:

Biomarker validation: Confirm findings in large, diverse patient populations.

Development of targeted assays: Create standardized LC-MS or immunoassay-based tests for specific lipids such as ceramides.

Diagnostic panel creation: Combine multiple lipid biomarkers into a predictive metabolic syndrome panel.

Clinical integration: Use lipid signatures alongside glucose, cholesterol, and BMI measurements to improve early detection and risk stratification.

Personalized medicine: Lipid profiles may guide treatment decisions, dietary interventions, and monitoring of therapeutic response.

Such lipid-based diagnostic panels could enable earlier detection of metabolic syndrome before severe complications like type 2 diabetes or cardiovascular disease develop.